SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes
Total Marks:20

QUESTIONS: 2

65/100

Annexure A

Descriptive Written Test

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

Cm-1 - AI-3

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(1) Intelligent agents in Smart Home automation

an intelligent agent is a system that perceives its environment through sensors and takes suitable action through actuators to achieve a desired objective.

* In a smart home, the agent can receive inputs such as motion, temperature, light level

a) Simple Reflex agent:

A simple reflex agent makes decision based only on the current percept using condition-action rules

Examples:

* If motion is detected -> turn on the light

* If temperature is high

* If no motion is detected.

Goal - based agent:

A goal based agent is one that selects actions based on how far they are from achieving a specific goal

Example:

If the goal is to maintain a comfortable room temperature while saving energy, the agent can decide whether to use the fan, AC

Best approach:

A hybrid agent combining Model - based goal based and utility based is better UCS can find the best cost

Conclusion:

For a maze where all movement have equal cost, IDS is preferable cost

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(2) UCS and IDS for ROBOT MAZE

Problem:

A robot placed in an unknown maze can use uninformed search strategies to explore possible paths find find the exit.

Advantage

* Complete when a step out are possible
* Optimal because it find the least - cost solution
* Useful when different paths have different cost.

Disadvantages

* Can require a large amount of memory
* Can be slow when there are many possible paths
* Time and space complexity can become very high.

Form example:

Depth 0 → Depth - 1 → Depth - 2 →

Depth 2 →

Two important methods are uniformed search strategies to explore possible paths and find the exit.

Two important methods are uniform cost search (UCS) and

Uniform cost search:

A robot placed in a maze can use uniformed search strategies to explore possible path and find the exit.

* Two important methods are uniform cost search.

For example:

* If a different maze path have different movement cost

* UCS will always explore the path with the lowest accumulated cost first.

Conclusion :

For a maze where all movement have equal cost, IDS is preferable.

⇒ Thus the agent type and search strategy should be selected according to the maze environment and robot objective.